BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. A task requires another task's result.

Q2. Amount of completed work per unit time.

Q3. Result depends on unsafe timing/order of shared operations.

Q4. D flip-flop Q=0, D=1 before next rising edge: what becomes Q?

### You Do pages · Independent questions

Q1. Sequential 11s [2]; parallel max7,4=7s [2].

Q2. Independent image load and help-page download [2]; validate/authorise then commit payment/order sequential [2], justify dependencies [2].

Q3. Both read same old value [2], both write overlapping update [1], one increment lost [1], lock/atomic increment [2].

### Exit Ticket · Model answer

False. Dependencies and overhead constrain speedup.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**





BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

# LESSON 064 · Lesson 064 · Thinking Concurrently: Independent Work

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is a process dependency?

Q2. What is throughput?

Q3. What is a race condition?

Q4. 031

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## I Do · Read new knowledge

### Key vocabulary

concurrency — Multiple tasks make progress in overlapping period.

parallelism — Actual simultaneous execution.

critical path — Longest dependency chain.

race condition — Incorrect timing-dependent shared state.

### New learning

Identify — Find tasks that do not depend on each other's intermediate results. Independent work can overlap.

Model — Distinguish concurrency (progress interleaved) from parallelism (truly simultaneous hardware execution).

Evaluate — Evaluate scheduling overhead and shared-resource/race risks, and explain why dependencies limit possible speedup.

### Worked example

Calculate minimum ideal time for independent image operations 6s and 4s on separate cores.

1. Explain the decision — Only independent tasks can safely overlap without waiting for each other.

## We Do · Practise with guidance

Task B takes 3s but depends on Task A's output (A takes 5s). Can separate cores shorten ideal 8s chain?

```
No: B cannot begin until A completes, so critical path is 5+3=8s.
```

Explain the decision

Extra processors do not remove logical dependency.





BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Independent jobs take 7s and 4s. Calculate sequential and ideal parallel completion times. [4]

### Question 2

Give two potentially concurrent stages of a school app and two stages that must be sequential. [6]

### Question 3

Two threads increment shared counter via read, add, write and occasionally lose updates. Explain race and one prevention. [6]

## Exit Ticket · Answer before checking

Doubling the number of CPU cores always halves task completion time.